

USER MANUAL

SPARKGANTT

A Gantt/schedule chart component for Ignition Perspective, virtualized to stay smooth past 200 tasks.

— VERSION 2.7.0

IGNITION 8.1.19+ AND 8.3

RUNS ON: ANY GATEWAY (PERSPECTIVE)

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1. OVERVIEW

SparkGantt is a Perspective component that renders a Gantt/schedule chart bound to tasks, groups, and dependency data from a dataset or from bindings. It virtualizes rows, labels, and dependency-arrow rendering once a view crosses 200 tasks, so scrolling and zooming stay responsive on large schedules. One source tree builds separate lines for Ignition 8.1 and 8.3.

SparkGantt is a display and interaction component, not a scheduling engine: it does not compute critical path, resource leveling, or auto-scheduling. It renders the tasks, groups, and dependencies you bind to it and reports drag/resize interactions back through its properties.

- Ignition 8.1 or 8.3, with the Perspective module enabled (separate build lines from one source tree).
- No other SparkBridge module is required. SparkGantt is a standalone product with its own license.

3. INSTALL

1. Download the signed SparkGantt .modl matching your Ignition line (8.1 or 8.3).
2. Gateway web UI > Config > Modules > Install or Upgrade a Module, upload the file, accept the EULA (see `license.html` at the repository root). Module id `com.sparkbridge.sparkgantt`.
3. If self-signed, accept the untrusted-certificate warning to complete install.

4. The component appears in the Designer Perspective palette after install; no build step or restart beyond the module install itself is required.

Soft-licensed the same way as other SparkBridge-suite modules, keyed to entitlement `com.sparkbridge.sparkgantt`. Drop `sparkbridge-license.key` in the gateway's data directory; the module re-reads it within 30 seconds. A missing license key only labels status surfaces and does not gate use. A present-but-invalid or tampered key fails startup closed and the component is not registered. Standalone paid product, sold and licensed on its own, never bundled into SparkBridge itself.

5. CONFIGURE

SparkGantt is configured entirely in the Ignition Designer property panel; there is no gateway config page.

PROPERTY	Meaning
tasks	Bound task list or dataset. Schema per task: id, label, start, end, progress (0-1), group, color, dependencies, milestone, tooltip.
groups	Bound group list; each group needs id and label.
datasetMapping	Maps dataset columns to the task schema fields when binding a raw dataset instead of an object array.

range	Visible time range: start, end, auto.
scale	Time-axis scale: auto, hour, day, week, month.
rowHeight	Pixel row height; clamped at runtime to a bounded min/max regardless of what a binding supplies.
showToday	Shows today's date marker.
nowLine	Shows a live now-line. Not independently controllable from showToday in the current release.
readOnly	Disables drag/resize interaction when true.
selection	Bound selected task id.
style	Style tokens; dark-theme aware.

A malformed row (missing mapping column, invalid id, non-finite date, duplicate id) is not silently dropped: it is excluded from the chart and reported in a data-quality banner (`.sparkganttt-data-quality-banner`) showing a count with the first 20 reasons in a tooltip. An interval with end before start is rejected outright rather than clamped to zero duration.

Day to day, watch the data-quality banner on any view whose task binding comes from a live dataset; a nonzero issue count means some rows are not rendering. Virtualization engages automatically above 200 tasks; below that threshold every

row, label, and dependency arrow renders in full. Drag and resize interactions write back through the bound task properties when `readOnly` is false.

7. VERIFY

- Place the component in a Perspective view, bind tasks and groups, and confirm the chart renders with no data-quality banner.
- Load a 200+ task dataset and confirm scrolling stays smooth; this is the point virtualization engages.
- Drag a task bar (with `readOnly` false) and confirm the bound start/end update.

An optional status route exists for licensing telemetry; it is soft and never gates chart functionality.

SYMPTOM	CAUSE	FIX
	Missing mapping column, invalid id, non-finite date, or duplicate id in the bound dataset	Open the banner tooltip for the first 20 reasons and fix the source rows; the chart still renders the valid rows
	Reversed interval is rejected, not clamped	Fix the source data so end is not before start

	Wrong build line installed for the gateway's Ignition version	Reinstall the .modl matching 8.1 or 8.3
	License key present but invalid or tampered	Replace sparkbridge-license.key with a valid key
	Dataset under the 200-task virtualization threshold but very complex bindings, or a very old build without the virtualization fixes	Upgrade to 2.7.0 or later; virtualization covers rows, labels, and dependency arrows uniformly above 200 tasks

9. REFERENCE

ITEM	Value
MODULE ID	com.sparkbridge.sparkgantt
LICENSE KEY PATH	<data>/sparkbridge-license.key, re-read within 30 s
VIRTUALIZATION THRESHOLD	Engages above 200 tasks (rows, labels, and dependency arrows)
TASK SCHEMA	id, label, start, end, progress, group, color, dependencies, milestone, tooltip
BUILD LINES	Ignition 8.1 and 8.3, one source tree

[SparkGantt install and config reference](#) · [Contact](#) · [Downloads and release notes](#)